Technical review

Item #1 · backdated-revision.pdf · Run ID: 20261113T232756Z-94fc69

Path	`R:\evidence\pdf\backdated-revision.pdf`
Size (bytes)	3613
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.pdf
Detected type (signature)	PDF document
Status	Needs attention
Case	RV-SAMPLE-001
Evidence ID	E0018
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`e51612fd222427bc85fa0f2a6b123b634355089087c1f63015174278b58aa34f`
md5	`e7aedb4ea7ff49b02133b26285c29b28`

Findings

[Medium] Different software between revisions (PRODUCER_CHANGED_BETWEEN_REVISIONS)

What it shows	The producer changes from “ReportLab PDF Library - (opensource)” in revision 1 to “Foxit PDF Editor Version 13.0.0” in revision 2.
What it does not prove	Documents legitimately pass through several programs (for example a signing or form tool); this shows a change of tool, not who used it or why.

[Medium] Revision dated before the previous one (REVISION_DATE_NOT_MONOTONIC)

What it shows	Revision 3 was appended to the file after revision 2, but its modification date (2026-03-03T14:15:00Z) is 1 d 3 h 0 min earlier than that of revision 2 (2026-03-04T17:15:00Z).
What it does not prove	Metadata dates come from the clock of the computer that saved each revision and can be set freely; a wrong clock, a time-zone error or dates copied by an editor can cause this. It does not by itself show deliberate backdating.

[Medium] PDF editor fingerprint (EDITOR_FINGERPRINT)

What it shows	Revision 2 carries the fingerprint of a PDF editor: Foxit PDF Editor / PhantomPDF (metadata producer: “Foxit PDF Editor Version 13.0.0”).
What it does not prove	Editors are also used for legitimate tasks such as filling in, signing, rotating or compressing a document; the fingerprint shows which software saved the file, not what was changed.

[Medium] Font added in a later revision (FONT_ADDED_IN_LATER_REVISION)

What it shows	Font Courier-Bold first appears in revision 3 and is used by page content there.
What it does not prove	Editing tools add fonts when a form is filled in, a stamp or comment is added or text is legitimately corrected; this shows that text was written in that revision, not that it falsifies the document.

[Low] Incremental updates (INCREMENTAL_UPDATES)

What it shows	The file contains 3 revisions: the original save plus later updates appended to the end of the file. Each earlier version can be extracted and compared.
What it does not prove	Incremental saving is normal when a PDF is filled in, signed, annotated or re-saved by many programs; it does not show who changed the file or that a change was improper.

[Informational] Time gap between revisions (REVISION_TIME_GAP)

What it shows	Revision 2 is dated 2 d 3 h 0 min after revision 1 (according to their modification dates).
What it does not prove	Metadata dates are written by the editing software from the computer clock and can be set freely; the gap does not prove when the changes were really made or by whom.

[Medium] Visible change between revisions (VISUAL_CHANGE_BETWEEN_REVISIONS)

What it shows	Rendering revision 1 and revision 2 gives different images on page(s) 1 (largest changed share 0.000058); pages added: -; pages removed: -. The changed areas are outlined in the comparison images.
What it does not prove	Incremental updates are normal when forms are filled, documents are signed or comments are added. It does not show who made the change, when, or with what intent.

[Medium] Visible change between revisions (VISUAL_CHANGE_BETWEEN_REVISIONS)

What it shows	Rendering revision 2 and revision 3 gives different images on page(s) 1 (largest changed share 0.001099); pages added: -; pages removed: -. The changed areas are outlined in the comparison images.
What it does not prove	Incremental updates are normal when forms are filled, documents are signed or comments are added. It does not show who made the change, when, or with what intent.

Revision history

#	Byte range	Saved at (Info)	Saved at (XMP)	Producer	Time since previous	Objects added / changed / deleted	Pages changed	Signed by (field)
1	0–1646 (1646 B)	2026-03-02T14:15:00Z	—	ReportLab PDF Library - (opensource)	—	8 / 0 / 0	—	—
2	1646–2621 (975 B)	2026-03-04T17:15:00Z	—	Foxit PDF Editor Version 13.0.0	2 days	1 / 2 / 0	1	—
3	2621–3613 (992 B)	2026-03-03T14:15:00Z	—	Foxit PDF Editor Version 13.0.0	?1 day	2 / 2 / 0	1	—

Timeline

#	Saved at (UTC)	Time since previous	Producer
1	2026-03-02 14:15:00	—	ReportLab PDF Library - (opensource)
2	2026-03-04 17:15:00	2 days	Foxit PDF Editor Version 13.0.0
3	2026-03-03 14:15:00	?1 day	Foxit PDF Editor Version 13.0.0

Dates are those stored in the file (Info dictionary, else XMP); the software that saved it can set them freely. Without usable dates the revisions are spaced evenly.

Signatures

None

The signer certificate, its chain and its revocation status were not validated: a mathematically valid signature does not show who controlled the key.

Fonts

Font	Subset prefix	Embedded	First revision	Object
Helvetica	—	no	1	2 0
Helvetica-Bold	—	no	1	3 0
Courier-Bold	—	no	3	10 0

Producer fingerprints

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	2026.10.21	37	loaded	`0b8df17b3ba7f33932cfc072932f198cdfdba98ad1332b33ac84c5c2f9068795`
local	—	—	—	absent	—

#	Tool	Category	Matched on	Value found	Traits not matching
1	ReportLab PDF Library	pdf_library	producer	ReportLab PDF Library - (opensource)	—
2	Foxit PDF Editor / PhantomPDF	pdf_editor	producer	Foxit PDF Editor Version 13.0.0	—

Visual comparison

Status	Compared
Pages changed	2

Difference images (in this item's folder):

- visual/rev-01-vs-02-page-001.png
- visual/rev-02-vs-03-page-001.png

Per-page measurements

Revisions	Page	Changed	Changed share	Changed pixels	Regions	DPI	Page size changed	Image / reason
1?2	1	yes	0.0058 %	66	1	110	no	visual/rev-01-vs-02-page-001.png
2?3	1	yes	0.1099 %	1244	1	110	no	visual/rev-02-vs-03-page-001.png

Parser warnings

None

The complete list of parser warnings is in warnings.json in this item's folder (its digest is in the audit log and in the extracted files table).

Extracted files

warnings.json (parser_warnings, 123 B)	md5: 251720985f0ba1c888116538bc3da921 sha256: 9dfdca46175986eda2dbcd6912894e5d123f4f0d267ee0e0b018d8d96fc149ae
revisions/rev-01.pdf (revision, 1646 B)	md5: 3888271753ea795369e203baab8ac540 sha256: 48d340c09a7c6d8cfd3a6f963f56272366b26d3e983d6c1e7a5cce06fa29caa1
revisions/rev-02.pdf (revision, 2621 B)	md5: dabe107d30c9e9f4da6245fb1c6e6b71 sha256: 2fd584534f39164bedfc6cdbfd9b5ed2622724f16e4da40d9481917a88046943
revisions/rev-03.pdf (revision, 3613 B)	md5: e7aedb4ea7ff49b02133b26285c29b28 sha256: e51612fd222427bc85fa0f2a6b123b634355089087c1f63015174278b58aa34f
objects.json (objects_dump, 6231 B)	md5: 69f4e6184986a6989b3b460371950eaa sha256: cd98994fb4b221843651135e0596fa62c6c3d59fd3945c696ec6de580950c1ea
visual/rev-01-vs-02-page-001.png (visual_diff, 29099 B)	md5: 2c6e72d0983fa593d65201bc39930931 sha256: e433ee948340ccd06b97aaaa89adf42add34bbcb17f75874e216d4a59ebafa81
visual/rev-02-vs-03-page-001.png (visual_diff, 31747 B)	md5: 23db79c1f9c761542663b4e33c2eb96c sha256: 9f82772a7220a19f03ff05d36fc001eab50e0bc15d31735d8b3dad49aeb82237

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms. PDF files were then parsed without being modified: every saved revision, its dates and producer, digital signatures, fonts and producer fingerprints were recorded, and pages of consecutive revisions were compared visually. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#5 – #16
Log folder	`R:\case\runs\rv-pdfauth\20261113T232756Z-94fc69\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- The absence of findings does not prove that a PDF is authentic or unaltered: a document can be forged before it is saved, or re-created so that no trace remains.
- When a PDF is printed to PDF, flattened, optimised or regenerated, its earlier revisions and most editing traces are lost; only its last state can be examined.
- Producer fingerprints come from a limited database (vendor documentation and files generated by Rakvor); producer strings can be typed by hand, and unknown or spoofed values cannot be recognised.
- Digital signatures are checked for byte coverage and cryptographic integrity only; certificate chains, revocation

and the signer's identity are not validated.
- The visual comparison between revisions renders only a limited number of pages per revision; pages beyond that limit are compared structurally only.